

AWS Architecture Review Packet

Field	Value
Student	Siarhei Hancharou
Course	CS 79D - Security in Amazon Web Services
Professor	Koda Kol
Project	AWS Learning Portal / AWSAwesome
Production application	https://awsawesome.com
GitHub repository	https://github.com/sergehall/final-project-CS79D
Deliverable selected	Option 3 - AWS Management Console review access
Prepared date	June 10, 2026

1. Submission Package

This document is the main final report for Option 3. The AWS Console credentials and SSH private key are intentionally not embedded in this report. They should be provided privately to the professor as separate files.

File	Purpose
Security_in_Amazon_Web_Services.txt	Contains the AWS Console sign-in URL, IAM review username, and temporary password for instructor review.
cs79d-professor-review-2026.pem	Optional SSH private key for terminal access to the EC2 instance if direct server inspection is requested.

2. Review Access

Item	Value
Deliverable selected	Option 3 - secure AWS Management Console review access with documentation
AWS account ID	960002046733
AWS Console sign-in URL	https://960002046733.signin.aws.amazon.com/console
Reviewer IAM user	koda-kol-cs79d-review
Temporary password	Provided privately in Security_in_Amazon_Web_Services.txt
Recommended access scope	Read-only console access for project resources during grading
Cleanup after grading	Disable/delete temporary IAM user and remove temporary SSH key access

SSH item	Value
Purpose	Optional EC2 Linux terminal inspection only; this does not grant AWS Console browser login.
Private key file	cs79d-professor-review-2026.pem
EC2 username	ec2-user
EC2 host	3.143.66.6
Professor command	ssh -i cs79d-professor-review-2026.pem ec2-user@3.143.66.6
Key fingerprint	SHA256:WBDRbJ3FtyWUmeWq5Ho6amuGIStO+swNeQ80shvFKSA

3. Verified Live Evidence

The following items were verified from the live EC2 instance and public HTTPS endpoint on June 10, 2026 UTC. Broad EC2, ALB, and Auto Scaling Describe permissions were intentionally unavailable to the application IAM role, which confirms that the runtime role is scoped to application needs instead of broad account inventory permissions.

Area	Verified result
EC2 host	SSH connected to ip-172-31-40-28.us-east-2.compute.internal; instance metadata returned i-018ad4a4d7c813990.
Runtime processes	PM2 shows cs79d-frontend and cs79d-backend online.
Network ports	Nginx listens on 80/443; Next.js listens on 3000; NestJS listens on 3001.
Health check	https://awsawesome.com/health returned HTTP 200 and JSON status ok.
IAM role	STS identity: arn:aws:sts::960002046733:assumed-role/cs79d-ec2-role/i-018ad4a4d7c813990.
S3	s3api head-bucket confirmed cs79d-uploads in us-east-2.
DynamoDB	scan on cs79d-activity-logs returned records from the live activity stream.
Lambda	lambda invoke of cs79d-log-action returned StatusCode 200.
API Gateway	App endpoint /api/aws-demo/invoke-lambda returned via api-gateway with gatewayName cs79d-gateway.
AI integration	/api/ai/security-advisor returned mode bedrock using amazon.nova-micro-v1:0.
Security headers	CSP, HSTS, X-Content-Type-Options, X-Frame-Options, and Referrer-Policy were present on https://awsawesome.com .
MDN Observatory	Scan id 99924955 at 2026-06-10T00:34:25Z returned A+, score 120, 10/10 tests passed.

4. Public Project Links

Production app: <https://awsawesome.com>

Architecture page: <https://awsawesome.com/architecture>

Final assessment / AI advisor page: <https://awsawesome.com/final-assessment-cp-practice-exam>

Live AWS evidence: <https://awsawesome.com/live-aws>

API demo: <https://awsawesome.com/api-demo>

Uploads workflow: <https://awsawesome.com/uploads>

Activity logs: <https://awsawesome.com/logs>

Cabinet security: <https://awsawesome.com/cabinet/security>

Lab 6 CloudFront/WAF evidence: <https://awsawesome.com/modules/lab-6-cloudfront-waf>

MDN Observatory scan: <https://developer.mozilla.org/en-US/observatory/analyze?host=awsawesome.com>

GitHub Pages slideshow: <https://sergehall.github.io/final-project-CS79D/>

Walkthrough video page: <https://sergehall.github.io/final-project-CS79D/video.html>

Amazon Bedrock: <https://awsawesome.com/api/ai/security-advisor>

5. Architecture Overview

AWSAwesome is a production-style AWS learning portal built as a full-stack cloud application. It includes a public Next.js frontend, a NestJS REST API backend, EC2-based application compute, Lambda serverless processing, S3 file storage, DynamoDB activity logging, CloudWatch monitoring, IAM roles, HTTPS/TLS protection, security headers, and Amazon Bedrock AI integration.

The live deployment uses a cost-controlled EC2 instance with Nginx as a reverse proxy so the project remains stable and accessible for grading. The broader production architecture also includes an Application Load Balancer and Auto Scaling Group design, which demonstrates how the same application can scale across multiple instances and maintain higher availability in a real-world AWS environment.

Layer	Implementation
Frontend	Next.js 16, React 19, TypeScript, Tailwind CSS; public routes for modules, uploads, architecture, API demo, and cabinet pages.
Backend	NestJS API on EC2, public proxy under <code>/api/*</code> , REST API with health, modules, uploads, logs, auth, MFA, OAuth, Lambda demo, and AI advisor endpoints.
Live compute	EC2 instance i-018ad4a4d7c813990 using PM2 for cs79d-frontend and cs79d-backend.
Serverless compute	AWS Lambda function cs79d-log-action invoked directly and through API Gateway.
Persistent storage	S3 bucket cs79d-uploads and DynamoDB table cs79d-activity-logs.
Security	IAM roles, security groups, HTTPS/TLS, HSTS, CSP, JWT, bcrypt, TOTP MFA, OAuth state validation, Turnstile support, throttling, upload validation, WAF lab.
Monitoring	CloudWatch dashboard, CPU alarm, Lambda logs, app /health endpoint, and activity logs.
AI	CloudWatch dashboard/alarm/logs and Amazon Bedrock model amazon.nova-micro-v1:0

6. AWS Services and Exact Resource Names

Service	Resource name / value	Purpose
EC2	cs79d-backend; i-018ad4a4d7c813990; Elastic IP 3.143.66.6	Runs the live frontend and backend application processes.
IAM	cs79d-ec2-role	Allows EC2 to access S3, DynamoDB, Lambda, CloudWatch, and Bedrock without hardcoded AWS keys.
S3	cs79d-uploads	Persistent validated upload storage for PDF, DOCX, JPG, JPEG, PNG, and TXT files.
DynamoDB	cs79d-activity-logs	Persistent activity log records for uploads, login/logout, Lambda demo, and AI events.
Lambda	cs79d-log-action	Serverless logging function that writes demo events to DynamoDB.
API Gateway	cs79d-gateway; POST /log-action; stage prod	HTTP API path used by the app to invoke the Lambda workflow.
Amazon Bedrock	amazon.nova-micro-v1:0 in us-east-1	AI Security Advisor for final-assessment recommendations.
CloudWatch	/aws/lambda/cs79d-log-action TargetTracking-cs79d-app-asg-AlarmHigh-8d2d5856-ae2c-4dcd-b3a6-d519389afbfc TargetTracking-cs79d-app-asg-AlarmLow-7c534a6b-a764-492c-8ec6-fee4baead43	Stores Lambda execution logs and provides Auto Scaling CPU monitoring through target tracking alarms.
Nginx / TLS	awsawesome.com and www.awsawesome.com	HTTPS reverse proxy to frontend port 3000 and backend port 3001.
CloudFront / AWS WAF	E2IJWFXTFQSGI3; d5fu4kwby0ghd.cloudfront.net; cs79d-lab6-cloudfront-web-acl	Lab 6 CDN and edge protection with IP, country, and user-agent block rules.

7. Production CloudFormation Topology

The production infrastructure template is in `infra/cloudformation/app-stack.yml`. It defines the scalable review topology with VPC networking, an Application Load Balancer, target groups, launch template, and Auto Scaling Group. The runtime IAM role does not have account-wide Describe permissions, so these names are documented from the project IaC and should be inspected in the professor's AWS Console review session.

Resource type	Name
VPC	vpc-0ed8d190686daed34
Internet Gateway	igw-0e3e95600bf606fe8
Public subnets	subnet-0134a2d073fc1762f
Route table	rtb-0fee188e15097f52e
ALB security group	cs79d-alb-sg
App security group	cs79d-backend-sg
Application Load Balancer	cs79d-alb
Frontend target group	cs79d-frontend-tg on port 3000, health check /
Backend target group	cs79d-backend-tg on port 3001, health check /health
Launch template	cs79d-app-template
Auto Scaling Group	cs79d-app-asg
ASG instance tag	cs79d-app-node
Scaling policy	cs79d-cpu-target-tracking; CPU target tracking, target 60 percent

8. Security Design

Security area	Implementation
Credential management	Application uses IAM roles instead of embedded AWS access keys. Reviewer credentials are temporary and shared separately.
Transport security	HTTPS with Let's Encrypt certificate on Nginx; HTTP redirects to HTTPS; HSTS max-age 31536000 with includeSubDomains and preload.
Browser security	Nonce-based CSP with default-src none, object-src none, frame-ancestors none, strict-dynamic, nosniff, X-Frame-Options, and strict referrer policy.
Authentication	JWT login, bcrypt password hashing, protected cabinet routes, password recovery flow, and optional Cloudflare Turnstile.
MFA	Authenticator app TOTP enrollment, verification, disable flow, and MFA challenge before issuing JWT.
OAuth	GitHub linked-provider flow with signed state values and environment-specific callback URLs.
Upload hardening	2 MB limit, extension/type allowlist, throttling, S3 persistence, and public access blocked on cs79d-uploads.
Abuse resistance	RequestThrottleGuard on uploads, sensitive auth routes, and AI advisor endpoint.
Network control	Security groups in the production topology restrict frontend/backend ports to ALB traffic; direct EC2 path uses Nginx as reverse proxy.
Edge security evidence	CloudFront/WAF lab blocks 130.166.0.0/16, IN/RU country traffic, and suspicious user agents such as sqlmap, nmap, curl, and BadBot.
External security scan	MDN Observatory A+, score 120, 10/10 tests passed on June 10, 2026 UTC.

9. Requirement Mapping

Assignment requirement	Evidence in this project
Frontend	Next.js public portal at https://awsawesome.com
Backend	NestJS API exposed through https://awsawesome.com/api/ routes and /health.
Compute	EC2 application host plus Lambda serverless logging function.
Two persistent storage services	S3 cs79d-uploads and DynamoDB cs79d-activity-logs.
At least five AWS services	EC2, IAM, S3, DynamoDB, Lambda, API Gateway, CloudWatch, Bedrock, CloudFront, WAF, ALB, Auto Scaling.
Auto Scaling and Load Balancing	CloudFormation topology defines cs79d-alb, target groups, launch template, and cs79d-app-asg.
Security	IAM roles, HTTPS/TLS, security headers, CSP, JWT, MFA, OAuth, Turnstile support, throttling, upload validation, WAF.
Monitoring and alerts	CloudWatch dashboard, CPU alarm, Lambda logs, /health endpoint, and DynamoDB activity logs.
AI integration	Amazon Bedrock AI Security Advisor using amazon.nova-micro-v1:0.
Architecture diagram	Included in this report and available at https://awsawesome.com/architecture .

10. Professor Review Checklist

1. Open <https://awsawesome.com> and confirm the application is live.
2. Open <https://awsawesome.com/health> and confirm JSON status ok.
3. Open /architecture and review how the services interact.
4. Open /final-assessment-cp-practice-exam and run the AI Security Advisor.
5. Open /api-demo and invoke the Lambda/API Gateway workflow.
6. Open /uploads and test a permitted small file upload.
7. Open /logs and confirm activity records are displayed from DynamoDB.
8. Sign in to AWS Console with the private credentials file and search for cs79d in us-east-2.
9. Inspect S3 cs79d-uploads, DynamoDB cs79d-activity-logs, Lambda cs79d-log-action, API Gateway cs79d-gateway, IAM roles, and CloudWatch logs.
10. Inspect CloudFront and WAF global resources for Lab 6 security evidence.
11. Review the MDN Observatory scan for awsawesome.com and confirm the A+ result.

11. Cost and Cleanup Plan

All services should remain active and accessible until grading is complete. After grading, cleanup should remove temporary access first and then tear down chargeable infrastructure.

Cleanup order	Action
1	Disable/delete the temporary reviewer IAM user.
2	Remove the temporary SSH public key from /home/ec2-user/.ssh/authorized_keys.
3	Empty and delete S3 buckets that are no longer needed.
4	Delete the CloudFormation app stack and Lab 6 CloudFront/WAF stack.
5	Confirm EC2, ALB, ASG, Lambda, API Gateway, DynamoDB, CloudWatch resources, and IAM roles are removed.
6	Release the Elastic IP if the domain is no longer serving the project.

Cost control	How it is addressed
Small compute	Course workload uses small EC2 instance types.
Auto Scaling bounds	Template uses low min/desired/max capacity settings for grading/demo mode.
DynamoDB billing	PAY_PER_REQUEST avoids provisioned capacity waste.
Upload cap	2 MB limit reduces storage and abuse risk.
Lambda	Short, event-driven executions for logging actions.
Bedrock	Amazon Nova Micro chosen as a low-cost model for AI review.